

Laboratory Exercise 10

Topics

1. Reading and writing files
2. Exception handling
3. Data processing

NOTE: Download also the .zip archive with the input text files to use in the different exercises!!

Discussion

- A. What happens if I try to read or write a file that doesn't exist?
- B. What is the difference between reporting and handling an exception?
- C. What is the difference between sequential access and random access?

Exercises (in red = suggested ones to complete during the lab)

Part 1 – Processing Files

Goal: For each of the following exercises, write a Python program that responds to the above requests. Complete at least one exercise during the lab session, and the remaining ones at home.

10.1.1 Enola Gay. Write a program that reads the text file `input.txt`, and writes each line read in the file `output.txt` preceded by its line number inserted between the characters `'/*'` and `'*/'`. For example, if the `input.txt` file contains:

```
Enola Gay
it is the bomber that, on August 6, 1945,
dropped the first atomic bomb on Hiroshima
nicknamed Little Boy.
```

The generated `output.txt` file must contain the text:

```
/*1*/Enola Gay
/*2*/is the bomber that, on August 6, 1945,
/*3*/dropped the first atomic bomb on Hiroshima
/*4*/nicknamed Little Boy.
```

[P7.2]

[SUGGESTED] 10.1.2 From the bottom. Write a program that reads all the lines of a `input.txt` text file, reverses their order, and writes them to another `output.txt` file. For example, if the `input.txt` file contains text:

```
Mary had a little lamb
Its fleece was white as snow
And everywhere that Mary went
The lamb was sure to go.
```

The `generated output.txt` file must contain the text:

```
The lamb was sure to go.  
And everywhere that Mary went  
Its fleece was white as snow  
Mary had a little lamb
```

[P7.9]

10.1.3 Searching for the Ring. Write a program that looks up a given word in the contents of a group of files. The program must ask the user to input the following:

- I. A list of **file names** on a single line, separated by commas;
- II. A word to look for.

File names must be split and stored in a list, while the search word must be stored in a variable. The program must display all lines that contain the word, even as a substring of other words, in a case insensitive way (ignoring differences between uppercase and lowercase). Each line that is displayed must be preceded by the name of the file in which it is located. For example, if the search word is `'ring'`, and the list contains the following files:

```
book.txt,address.txt,homework.py
```

then the program must display the lines where the word is found with the following format:

```
book.txt: There is only one Lord of the Ring, only one who can bend it to  
his will  
book.txt: The ring has awoken; it's heard its master's call.  
address.txt: Kris Kringle, North Pole  
address.txt: Homer Simpson, Springfield  
homework.py: string = "text"
```

[P7.6]

[SUGGESTED]10.1.4 In the hotel. The administrative manager of a hotel records sales in a text file. Each line contains the following 4 pieces of information, separated by semicolon characters (';'):

- I. the name of the customer;
- II. the service sold;
- III. the amount paid;
- IV. the date of the event.

The services sold can be, for example, a **dinner**, a **conference**, or **accommodation**. The correct format for this file is that it contains 4 fields per line, and that the amount paid contains `float` values. Write a program that reads this text file, and displays the total amount paid for each type of service, reporting an error if the file does not exist or if its format is incorrect. [P7.29]

10.1.5 Second Possibility. Write a program that prompts the user to enter a series of `float` values. When the user enters a value that is not of the correct type, the program must give the user a second chance to enter the value, and after two chances it must stop reading the input, immediately terminating the program. Data entry continues until the user enters an empty string ('') as input. Sum all the correctly specified values and display their sum when the user has finished entering the data. Use exception handling to catch improper input. [P7.13]

Part 2 – Data processing: correspondences and ciphers

Goal: For each of the following exercises, write a Python program that responds to the above requests. Complete at least one exercise during the lab session, and the remaining ones at home.

10.2.1 Random Monoalphabetic Encryption. The Caesar cipher is a monoalphabetic substitution cipher, in which each letter of the plaintext is replaced, in the ciphertext, by the letter that is a certain number of positions later in the alphabet. Since the number of positions used for translation is fixed, Caesar's cipher is too easy to break. To make the ciphertext more difficult to decipher, an alternative strategy is to use a word as a key instead of a number. For example, if the word you use as the key is FEATHER, first you delete the duplicate letters, getting FEATHR, and then add the remaining letters of the alphabet to the end, in reverse order. The following sequence is obtained.

F E A T H R Z Y X W V U S Q P O N M L K J I G D C B

Then, the letters are encrypted, according to the following scheme.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
F	E	A	T	H	R	Z	Y	X	W	V	U	S	Q	P	O	N	M	L	K	J	I	G	D	C	B

Any other characters (such as spaces, digits, or punctuation marks) must remain unchanged.

Write a program that asks the user for the name of a text file to be encrypted or decrypted, the choice of which operation to perform (encryption or decryption), a keyword, and the name of a file in which to write the output of the processing. The program must store the keyword entered by the user in a variable, and use it to encrypt or encrypt/decrypt the text file provided as input, then write the result of the processing in the indicated output file. [P7.20]

[SUGGESTED]10.2.2 University record book. Write a program that displays the list of exams passed by a student, with their grades. There is a `classes.txt` file, which contains the codes of all the courses taught at the educational institution (a US college), one per line, like this:

```
CSC1
CSC2
CSC46
CSC151
MTH121
...
```

For each course, there is a file whose name matches the course code, followed by `'.txt'`. This file lists the people who have passed the course exam, reporting for each person student ID and the grade obtained. For example, the `CSC2.txt` file might contain the text:

```
11234 A-
12547 B
16753 B+
```

21886 C

...

Write a program that requires the user to enter a 'Student ID' as input, and then displays the list of exams passed by the student, reporting the marks obtained for each exam, respecting the following format.

Student ID: 16753

CSC2 B+
 MTH121 C+
 CHN1 A
 PHY50 A-
 [P7.28]

10.2.3 Playfair Cipher. It is possible to decipher a text by simply analyzing the frequency of occurrence of individual letters. One way to hinder this strategy is to encode pairs of letters together. The Playfair cipher is a simple scheme that applies this strategy. In this scheme, the first thing you do is choose a keyword, and eliminate the duplicate letters in it. Then the keyword thus elaborated is inserted into a 5×5 chessboard, followed by the remaining letters of the English alphabet in order. Since there are only 25 squares and the English alphabet has 26 letters, the 'I' and 'J' are considered indistinguishable. Check that the letters in the source text are even, otherwise add a 'Z'. Using 'PLAYFAIR', transformed into 'PLAYFIR', as the keyword, you get the following pattern.

P	L	A	Y	F
I	R	B	C	D
E	G	H	K	M
N	O	Q	S	T
U	V	W	X	Z

Starting from this table, to encrypt a pair of letters, for example 'AM', we identify the portion of the table that has 'A' and 'M' at two extremes:

P	L	A	Y	F
I	R	B	C	D
E	G	H	K	M
N	O	Q	S	T
U	V	W	X	Z

The encryption of the 'AM' pair is obtained by finding the values at the two remaining extremes of the table portion, in this case, 'FH'. If the two letters of the pair are on the same row or on the same column, such as the 'GO' pair, the encryption is performed by exchanging the letters, obtaining, in this case, 'OG'. Decryption follows the same rules. Write a program that encrypts or decrypts a text file according to the Playfair cipher scheme. [P7.23]

10.2.4 Covalent bonds. Suppose a file contains the bond energies and lengths for covalent bonds, such as those shown in the following table.

Bond (single , double , or triple)	Bond energy (kJ/mol)	Bond length (nm)
C C	370	0.154
C C	680	0.13
C C	890	0.12
C H	435	0.11
C N	305	0.15
C O	360	0.14
C F	450	0.14
C Cl	340	0.18
O H	500	0.10
O Or	220	0.15
O Yes	375	0.16
N H	430	0.10
N O	250	0.12
F F	160	0.14
H H	435	0.074

The file format (called `bond_data.txt`) requires that each row in the table corresponds to a line of text, and that, in each row, the three fields are separated by a space (' '). The file does not include a header. For example, a file that contains the data presented in the table might contain the text:

```
C|C 370 0.154
C||C 680 0.13
C|||C 890 0.12
...
```

Write a program that takes data from one column as input and displays matching data from the other two columns in the file. If the input data matches multiple rows in the table, the program must return all of them. For example, a bond length of `0.12` must return:

- the triple bond `C|||C` and the bond energy `890 kJ/mol`,
- the single bond `N|O` and the bond energy `250 kJ/mol`.

[P7.34]